

Assignment 6

Problems:

1. Consider $\mathbb{R}^4$ with the standard inner product. Let W be the subspace of $\mathbb{R}^4$ consisting of all vectors which are orthogonal to both $\alpha = (1, 0, -1, 1)$ and $\beta = (2, 3, -1, 2)$. Find a basis for W .
2. Let V be an inner product space, and let α and β be vectors in V . Show that $\alpha = \beta$ if and only if $\langle \alpha | \gamma \rangle = \langle \beta | \gamma \rangle$ for every γ in V .

3. Let

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \quad \mathbf{p}_2 = \begin{bmatrix} 4 \\ -2 \\ -6 \\ -7 \end{bmatrix} \quad \mathbf{p}_3 = \begin{bmatrix} 3 \\ 4 \\ -2 \\ 1 \end{bmatrix}$$

and

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 7 \end{bmatrix}.$$

Determine the best approximation, $\hat{\mathbf{x}}$, of the vector $\mathbf{x}$ by vectors in $W = \text{span}[\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3]$. Also, determine the projection of $\mathbf{x}$ onto the orthogonal complement of W .

4. Let S be a subset of an inner product space V . Show that $(S^\perp)^\perp$ contains the subspace spanned by S . When V is finite-dimensional, show that $(S^\perp)^\perp$ is the subspace spanned by S .
5. Let V be a finite-dimensional inner product space, and let $\mathcal{B} = (\alpha_1, \dots, \alpha_n)$ be an orthonormal basis for V . Let T be a linear operator on V ; and A , the matrix of T in the ordered basis $\mathcal{B}$. Prove that

$$A_{ij} = \langle T\alpha_j | \alpha_i \rangle.$$

6. Suppose $V = W_1 \oplus W_2$ and that f_1 and f_2 are inner products on W_1 and W_2 , respectively. Show that there is a unique inner product f on V such that
 - (a) $W_2 = W_1^\perp$;
 - (b) $f(\alpha, \beta) = f_k(\alpha, \beta)$, when α, β are in $W_k, k = 1, 2$.

Optional Problems:

1. *Solution set of a quadratic inequality.* Let $C \subseteq \mathbf{R}^n$ be the solution set of a quadratic inequality,

$$C = \{x \in \mathbf{R}^n \mid x^T A x + b^T x + c \leq 0\},$$

with $A \in \mathbf{S}^n$, $b \in \mathbf{R}^n$, and $c \in \mathbf{R}$.

- (a) Show that C is convex if $A \succeq 0$.
(b) Show that the intersection of C and the hyperplane defined by $g^T x + h = 0$ (where $g \neq 0$) is convex if $A + \lambda g g^T \succeq 0$ for some $\lambda \in \mathbf{R}$.

Are the converses of these statements true?

2. Suppose $f : \mathbf{R} \rightarrow \mathbf{R}$ is convex, and $a, b \in \text{dom } f$ with $a < b$.

- (a) Show that

$$f(x) \leq \frac{b-x}{b-a} f(a) + \frac{x-a}{b-a} f(b)$$

for all $x \in [a, b]$.

- (b) Show that

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(a)}{b - a} \leq \frac{f(b) - f(x)}{b - x}$$

for all $x \in (a, b)$. Draw a sketch that illustrates this inequality.

- (c) Suppose f is differentiable. Use the result above to show that

$$f'(a) \leq \frac{f(b) - f(a)}{b - a} \leq f'(b).$$

- (d) Suppose f is twice differentiable. Use the result above to show that $f''(a) \geq 0$ and $f''(b) \geq 0$.

3. Show that a set is convex if and only if its intersection with any line is convex. Show that a set is affine if and only if its intersection with any line is affine.
4. *Midpoint convexity.* A set C is *midpoint convex* if whenever two points a, b are in C , the average or midpoint $(a + b)/2$ is in C . Obviously a convex set is midpoint convex. It can be proved that under mild conditions midpoint convexity implies convexity. As a simple case, prove that if C is closed and midpoint convex, then C is convex.